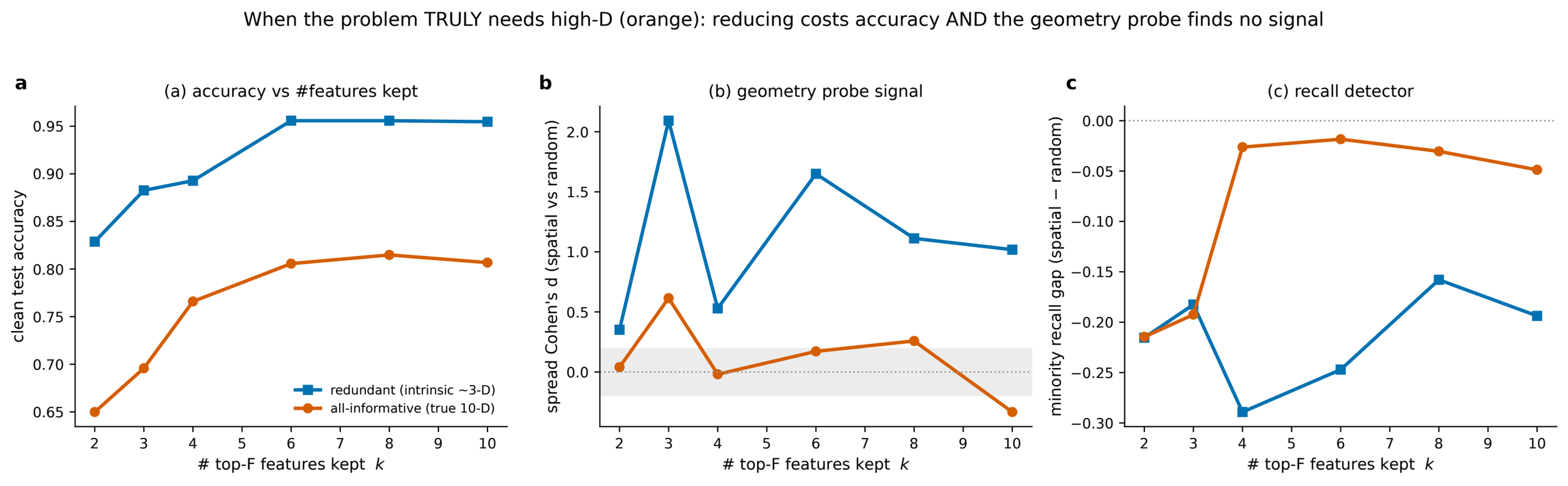
When the problem TRULY needs high-D (orange): reducing costs accuracy AND the geometry probe finds no signal
a
(a) accuracy vs #features kept
clean test accuracy
0.95
0.90
0.85
0.80
0.75
0.70
0.65
redundant (intrinsic ~3-D)
all-informative (true 10-D)
2
3
4
5
6
7
8
9
10
top-F features kept k
b
(b) geometry probe signal
spread Cohen's d (spatial vs random)
2.0
1.5
1.0
0.5
0.0
2
3
4
5
6
7
8
9
10
top-F features kept k
c
(c) recall detector
minority recall gap (spatial − random)
0.00
−0.05
−0.10
−0.15
−0.20
−0.25
−0.30
2
3
4
5
6
7
8
9
10
top-F features kept k